INCH-POUND

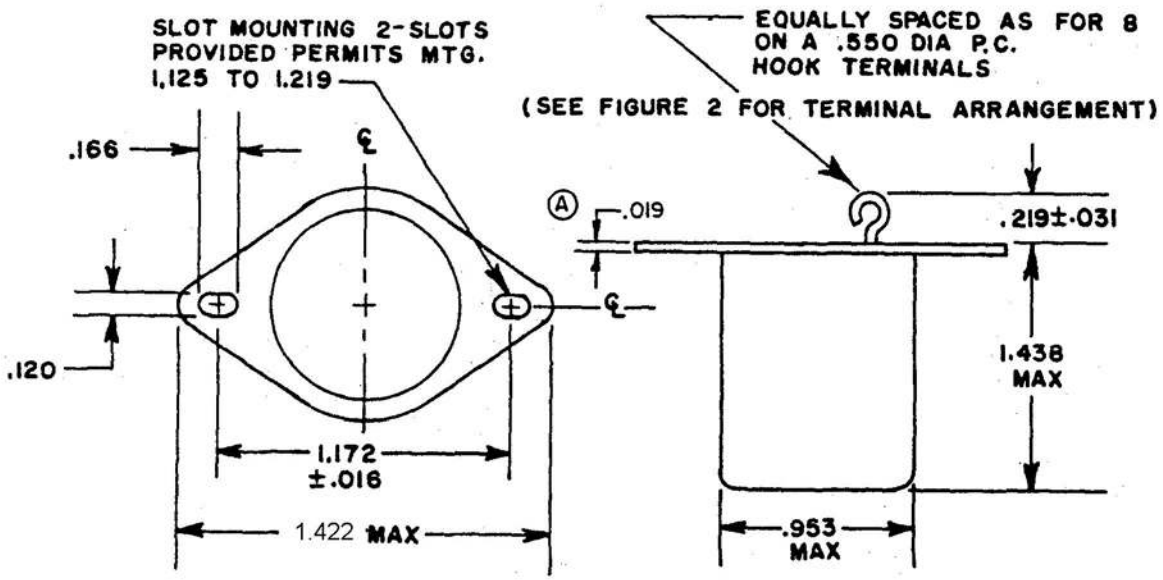
MIL-PRF-27/166B
18 July 2008
SUPERSEDING
MIL-PRF-27/166A
22 January 1985

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-27.




CIRCUIT DIAGRAM AND MARKING
(SEE TABLE I AND FIGURE 3)

| Inches | mm |
|---|---|
| .016 | 0.41 |
| .019 | 0.48 |
| .031 | 0.79 |
| .120 | 3.05 |
| .166 | 4.22 |
| .219 | 5.56 |
| .550 | 13.97 |
| .953 | 24.21 |
| 1.125 | 28.58 |
| 1.172 | 29.77 |
| 1.219 | 30.96 |
| 1.422 | 36.12 |
| 1.438 | 36.53 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.
4. Primary and secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.

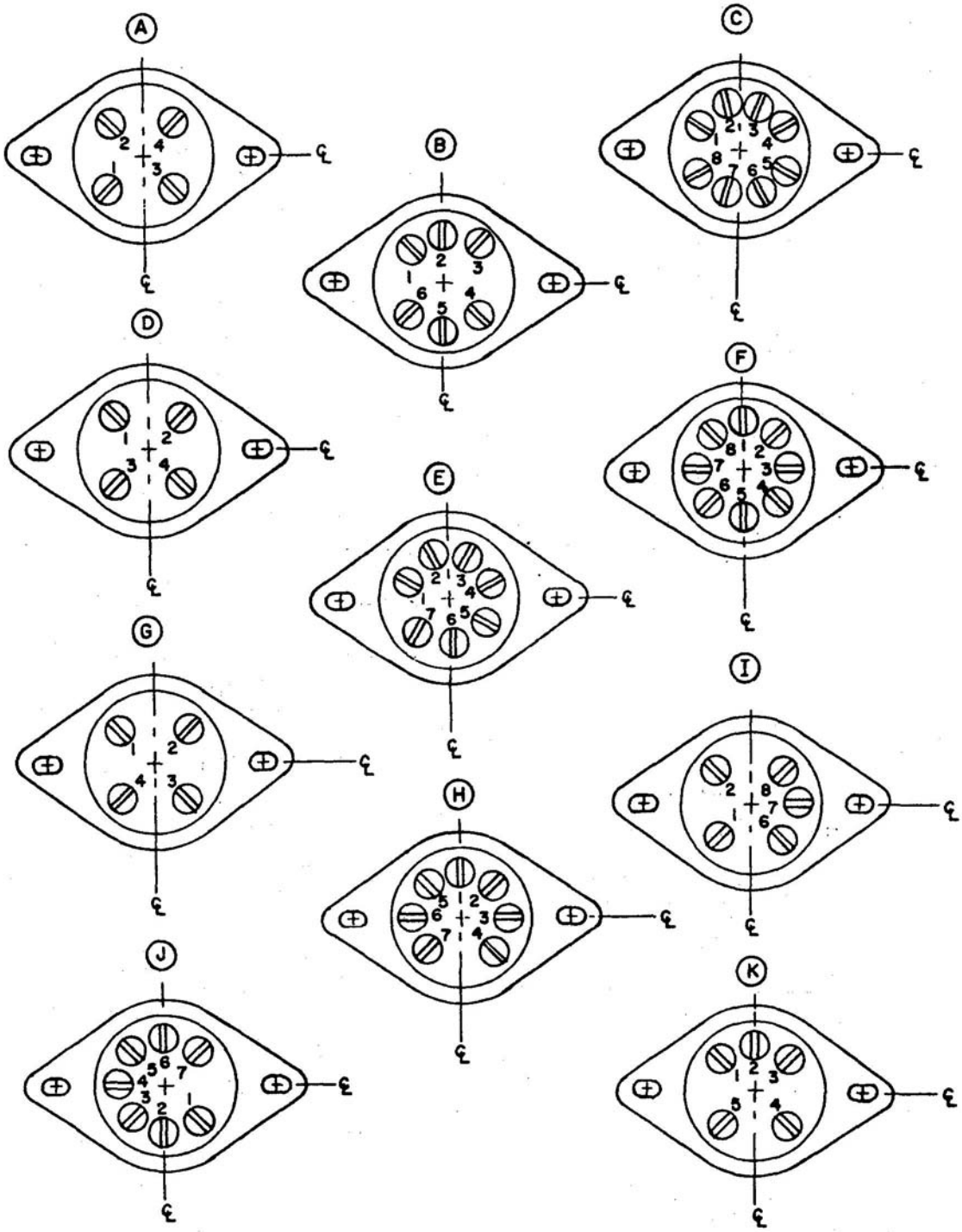



FIGURE 2. Terminal arrangement.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.

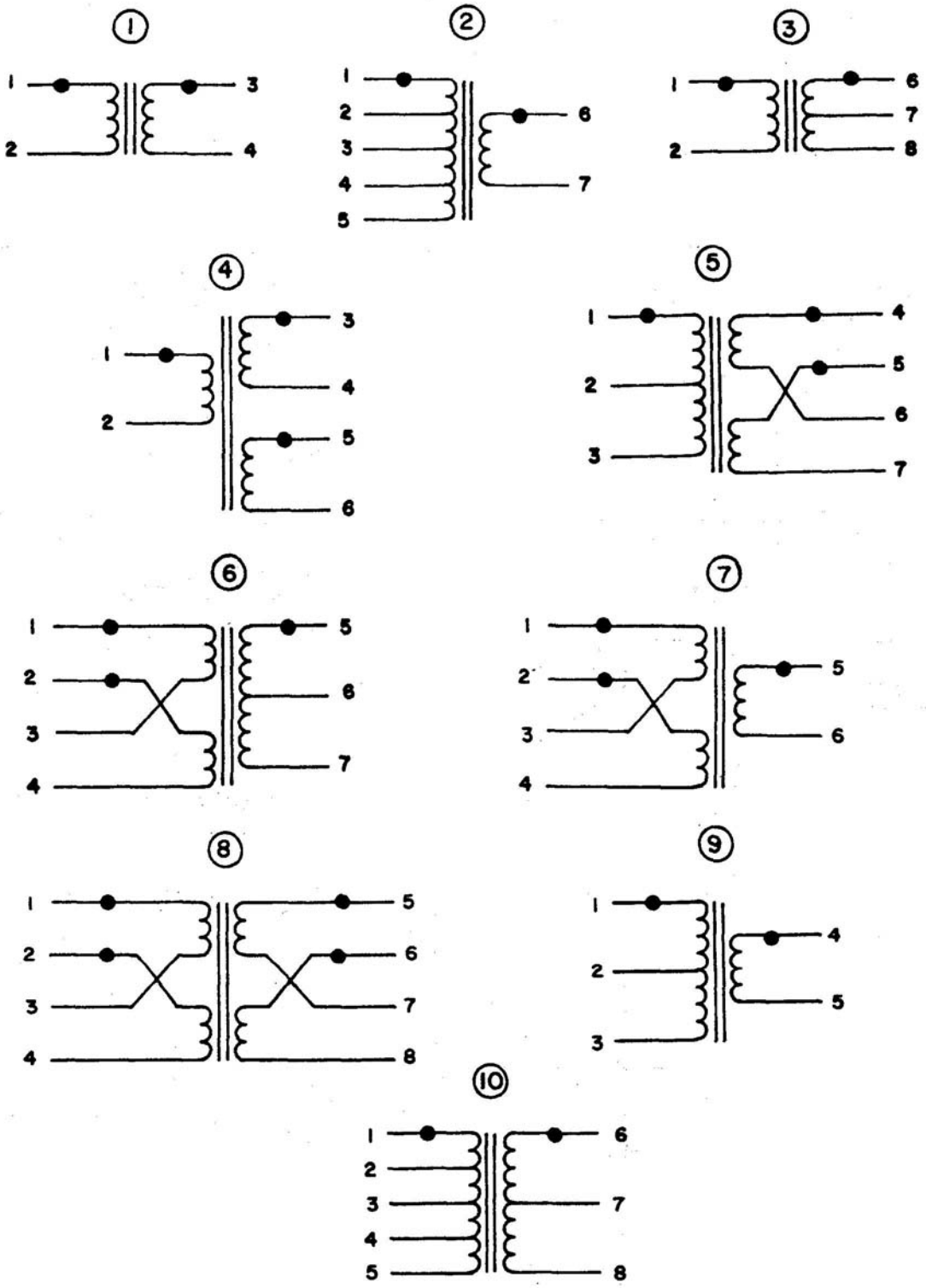



FIGURE 3. Circuit diagram.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical ratings. 4/

| Dash No. | Primary 1/ Impedance (ohms) | Secondary 1/ Impedance (ohms) | Primary DCR ±25% (ohms) | Secondary DCR ±25% (ohms) | Power Level (mW) | Primary dc current (mA) | Working voltage (volts) | Dielectric withstanding voltage 2/ | | Terminal arrangement (see figure 2) | Circuit diagram (see figure 3) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Sea level | Reduced Barometric pressure | | |
| 01 | 82 | 135 k | 14 | 6700 | 63 | 50 | 250 | 750 | 300 | G | 1 |
| 02 | 50, 200 CT<br>500 CT | 50 k | 54 | 3827 | 3 | 0 | 175 | 500 | 300 | H | 2 |
| 03 | 50, 200 CT<br>500 CT | 50 k CT | 54 | 3827 | 3 | 0 | 175 | 500 | 300 | F | 10 |
| 04 | 500/125<br>SPLIT | 150/37.5<br>SPLIT | 35 | 16 | 250 3/ | 20 | 175 | 500 | 300 | F | 8 |
| 05 | 500/125<br>SPLIT | 500/125<br>SPLIT | 37 | 67 | 250 3/ | 20 | 175 | 500 | 300 | F | 8 |
| 06 | 600/150<br>SPLIT | 600 CT | 39 | 29 | 6.3 | 0 | 175 | 500 | 300 | J | 6 |
| 07 | 600/150<br>SPLIT | 2k/500<br>SPLIT | 70 | 280 | 31 | 0 | 175 | 500 | 300 | F | 8 |
| 08 | 600/150<br>SPLIT | 1 M | 16.5 | 5100 | 10 | 0 | 175 | 500 | 300 | B | 7 |
| 09 | 1500 CT | 500/125<br>SPLIT | 100 | 35 | 100 | 8 | 175 | 500 | 300 | J | 5 |
| 10 | 2000 CT<br>4000 CT | 8<br>16 | 200 | 0.9 | 100 | 4 | 175 | 500 | 300 | K | 9 |
| 11 | 10 k CT | 500/125<br>SPLIT | 750 | 52 | 100 | 4 | 175 | 500 | 300 | E | 5 |
| 12 | 10k/2.5k<br>SPLIT | 2k/500<br>SPLIT | 780 | 195 | 100 | 4 | 175 | 500 | 300 | C | 8 |
| 13 | 10k/2.5k<br>SPLIT | 4k/1k<br>SPLIT | 800 | 350 | 100 | 4 | 175 | 500 | 300 | C | 1 |





Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical ratings - Continued. 4/

| Dash No. | Primary 1/ Impedance (ohms) | Secondary 1/ Impedance (ohms) | Primary DCR ±25% (ohms) | Secondary DCR ±25% (ohms) | Power Level (mW) | Primary dc current (mA) | Working voltage (volts) | Dielectric withstanding voltage 2/ | | Terminal arrangement (see figure 2) | Circuit diagram (see figure 3) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Sea level | Reduced Barometric pressure | | |
| 14 | 10 k | 1 M | 330 | 4875 | 10 | 0 | 175 | 500 | 300 | A | 1 |
| 15 | 15 k | 60 k | 690 | 3100 | 4 | 0 | 175 | 500 | 300 | A | 1 |
| 16 | 15 k | 60 k | 690 | 3100 | 25 | | 175 | 500 | 300 | D | 1 |
| 17 | 15 k | 95 k CT | 673 | 3800 | 2 | 0 | 175 | 500 | 300 | I | 3 |
| 18 | 15 k | 95 k SPLIT | 1260 | 5800 | 12 | 4 | 175 | 500 | 300 | B | 4 |
| 19 | 20 k CT | 600/150 SPLIT | 930 | 45 | 125 | 4 | 175 | 500 | 300 | J | 5 |

1/ Impedance values written one above the other indicate a range of matching impedances over which the parts will give satisfactory performance as long as the impedance ratio is maintained.
2/ Dielectric withstanding voltage between split sections, 50 volts rms maximum.
3/ 250 mW at 100 Hz, 1 watt at 200 Hz.
4/ Qualification testing and approval to M27/166-01 shall be sufficient to grant qualification approval to M27/166-02 through M27/166-19, inclusive.




Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Working voltage: See table I.

Frequency range: See table II.

Design and construction:

Dimensions and configurations: See figure 1.

Duty cycle: Continuous.

Case: Steel drawn, hermetically sealed.

Terminals: Hooked pin with compressed glass bead header.

Weight: 2.0 ounces maximum.

Altitude: 50,000 feet maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

Dielectric withstanding voltage: See table I.

Electrical characteristics: See table II.

Frequency response: See table II.

Reference frequency: 1 kHz.

Insertion loss: 3 dB at 1 kHz maximum.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/166-(dash number from table I).



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical characteristics.

| Dash no. | No load (center-tap Voltage unbalanced only) | | Frequency response | | | Resonance second resonant frequency (min) (kHz) | Polarity same on terminals (see figure 3) |
|---|---|---|---|---|---|---|---|
| | Potential at 1 kHz (volts) | Unbalanced ±% | Frequency range (Hz) | ± (dB) | $E_L$ (volts) | | |
| 01 | - - - | - - - | 250 – 8,000 | 2 | 92 | 8 | 1, 3 |
| 02 | 1.0 | 1.0 | 50 – 10,000 | 2 | 12.2 | 10 | 1, 6 |
| 03 | 1.0 | 1.0 | 50 – 10,000 | 2 | 12.2 | 10 | 1, 6 |
| 04 | 1.0 | 1.0 | 100 – 20,000 | 2 | 6.1 | 20 | 1, 2, 5, 6 |
| 05 | 1.0 | 1.0 | 100 – 20,000 | 2 | 11.2 | 20 | 1, 2, 5, 6 |
| 06 | 1.0 | 1.0 | 50 – 10,000 | 2 | 1.9 | 10 | 1, 2, 5 |
| 07 | 1.0 | 1.0 | 50 – 20,000 | 2 | 7.9 | 20 | 1, 2, 5, 6 |
| 08 | 1.0 | 1.0 | 200 – 3,000 | 4 | 100 | 3 | 1, 2, 5 |
| 09 | 1.0 | 1.0 | 100 – 10,000 | 2 | 7.1 | 10 | 1, 4, 5 |
| 10 | 1.0 | 1.0 | 100 – 10,000 | 2 | 0.9 | 10 | 1, 4 |
| 11 | 1.0 | 1.0 | 100 – 10,000 | 2 | 7.1 | 10 | 1, 4, 5 |
| 12 | 1.0 | 1.0 | 100 – 10,000 | 2 | 14 | 10 | 1, 2, 5, 6 |
| 13 | 1.0 | 1.0 | 100 – 10,000 | 2 | 20 | 10 | 1, 2, 5, 6 |
| 14 | - - - | - - - | 200 – 3,000 | 4 | 100 | 3 | 1, 3 |
| 15 | - - - | - - - | 50 – 10,000 | 2 | 15.5 | 10 | 1, 3 |
| 16 | - - - | - - - | 200 – 10,000 | 2 | 39 | 10 | 1, 3 |
| 17 | 1.0 | 1.0 | 50 – 10,000 | 2 | 17 | 10 | 1, 6 |
| 18 | 1.0 | 1.0 | 200 – 10,000 | 2 | 34 | 10 | 1, 3, 5 |
| 19 | 1.0 | 1.0 | 200 – 10,000 | 2 | 8.7 | 10 | 1, 4, 5 |

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2008-019)

Review activities:
Army – AR, CR4
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:43Z
Check the source to verify that this is the current version before use.